ACSC383 - User Requirements Specification

1. Account and Access Requirements

1. Users must be able to create an account using a valid university email address.
2. Users must be required to verify their email address before accessing the system.
3. Users must be able to securely log in and log out of the system.
4. Users must only be able to access system features after authentication.
5. Users must be able to reset their password if forgotten.

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2. User Profile Requirements

6. Users must have a personal profile displaying their basic account information.
7. Users must be able to view all items they have listed.
8. Users must be able to view their active, pending, and completed transactions.
9. Users must be able to view their cumulative environmental impact (CO_2 and waste saved).
10. Users must be able to edit limited profile details (e.g. display name).

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3. Item Listing Requirements

11. Users must be able to list an item for **swap**, **sale**, or **donation**.
12. Users must be required to provide a title, description, category, and condition for each item.
13. Users must be able to upload one or more images per item.
14. Users must be able to specify a price when listing an item for sale.
15. Users must be able to edit their own item listings.
16. Users must be able to delete their own item listings.
17. Users must be able to see the current status of their items (available, reserved, completed).

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4. Browsing and Discovery Requirements

18. Users must be able to browse all available items listed by other users.
19. Users must be able to search for items using keywords.
20. Users must be able to filter items by category.

21. Users must be able to filter items by condition.
22. Users must be able to filter items by transaction type (swap, sale, donation).
23. Users must only be shown items that are currently available.

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5. Item Details Requirements

24. Users must be able to view detailed information for each item.
25. Item detail views must display item images, description, condition, and transaction type.
26. Users must be able to view the item owner’s basic information.
27. Users must be able to view the estimated environmental impact of reusing the item.

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6. Swap Transaction Requirements

28. Users must be able to propose a swap for an available item.
29. Users must be able to select one of their own listed items when proposing a swap.
30. Users must not be able to propose a swap using an unavailable or reserved item.
31. Users must be notified when a swap request is received.
32. Users must be able to accept or decline a swap request.
33. Users must not be able to accept more than one transaction for the same item.
34. Users must have both items reserved when a swap is accepted.
35. Users must be able to cancel a pending or accepted swap before completion.

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7. Sale and Donation Transaction Requirements

36. Users must be able to request to buy an item listed for sale.
37. Users must be able to request an item listed for donation.
38. Item owners must be able to accept or decline sale or donation requests.
39. Accepted sale or donation requests must reserve the item.
40. Users must be able to cancel sale or donation transactions before completion.

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8. Transaction Completion Requirements

41. Users must be able to mark a transaction as completed.
42. Users must be able to upload a photo as proof of item receipt.
43. Users must not be able to complete a transaction without confirmation.

44. Completed items must no longer appear in browsing results.
45. Users must be able to view completed transactions in their history.

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9. Notifications and Feedback Requirements

46. Users must receive notifications for transaction proposals.
47. Users must receive notifications for acceptance or rejection of transactions.
48. Users must receive notifications when a transaction status changes.
49. Users must be able to view notifications within the system.

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10. Environmental Impact Requirements

50. Users must be able to see estimated CO_2 emissions saved per transaction.
51. Users must be able to see estimated waste diverted per transaction.
52. Users must be able to view a cumulative environmental impact summary.
53. Users must understand that environmental values are estimates, not exact measurements.

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11. Constraints and Usage Rules (User Perspective)

54. Users must only be able to interact with other verified university members.
55. Users must not be able to modify items or transactions they do not own.
56. Users must not be able to participate in multiple active transactions using the same item.
57. Users must rely on offline methods to exchange items physically.
58. Users must use the system responsibly and in accordance with acceptable use policies.

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12. Overall User Experience Requirements

59. Users must be able to use the system comfortably on a smartphone.
60. Users must be able to understand the system without training.
61. Users must be able to complete core actions (listing, browsing, swapping) with minimal steps.
62. Users must receive clear feedback after every important action.